

Cost Management — Easy KCNA Notes

1. What is Cost Management?

Cost Management means monitoring, controlling, and optimizing the money spent on cloud and Kubernetes resources.

In simple words:

 **Cost Management = Know where your money is going and avoid unnecessary spending.**

2. Why is Cost Management Important?

Cloud resources are usually charged based on usage.

For example, you may pay for:

-  Virtual machines / Nodes
-  Storage
-  Network traffic
-  Databases
-  Kubernetes workloads
-  Cloud services

If you don't monitor them, your bill can become unexpectedly high.

3. Kubernetes Cost Example

Imagine you have:

Kubernetes Cluster

|

|— Node 1 

|— Node 2 

└— Node 3 

But your application only needs two Nodes.

Node 1 → Used 

Node 2 → Used 

Node 3 → Mostly unused 

You're still paying for Node 3.

Cost management helps you identify this.

Unused resources



Identify



Reduce/remove



 Save money

4. Resource Requests and Limits

This connects with the **Resource Limits** topic you learned earlier.

For example:

resources:

requests:

cpu: "500m"

memory: "512Mi"

limits:

cpu: "1"

memory: "1Gi"

Requests

Tell Kubernetes:

"This container normally needs this amount of resources."

Limits

Tell Kubernetes:

"Don't allow this container to use more than this configured amount."

Properly setting requests and limits can help prevent **over-provisioning** and improve resource utilization.

5. Right-Sizing ★

What is Right-Sizing?

Right-sizing means giving workloads the appropriate amount of resources they actually need.

Example:

Application needs:

CPU = 500m

Memory = 512Mi

But you give it:

CPU = 4

Memory = 8Gi

That's potentially wasteful.

Instead:

CPU = 500m

Memory = 512Mi

is closer to the actual requirement.

🧠 **Remember:**

Right-sizing = Don't give too much, don't give too little.

6. Autoscaling and Cost Management ★

Autoscaling can help control costs by adjusting capacity according to demand.

High demand:

More workload



More Pods



More capacity

Low demand:

Less workload



Fewer Pods/Nodes



Less unused capacity

For example:

HPA → adjusts Pods 📦

Cluster Autoscaler → adjusts Nodes 🖥️

This can help avoid paying for unnecessary capacity.

7. Monitor Resource Usage 📊

You should monitor:

- CPU usage
- Memory usage
- Number of Pods
- Number of Nodes
- Storage usage
- Network usage

- Cloud-service costs

Example:

Node 1 → CPU 80% 

Node 2 → CPU 75% 

Node 3 → CPU 5% 

Node 3 may be a candidate for investigation.

8. Avoid Over-Provisioning

Over-provisioning means:

Giving more resources than the application actually needs.

Example:

Actual need:

CPU = 1 CPU

Memory = 1 GB

Provided:

CPU = 8 CPU

Memory = 16 GB

A lot of capacity is sitting unused.

Actual usage 

Allocated 

Better resource planning can reduce waste.

9. Cost Management in Cloud Native

Cloud Native environments often use:

Containers



Kubernetes



Cloud Infrastructure

Each layer can contribute to cost.

For example:

Cloud

|

├— Compute 

├— Storage 

├— Network 

└— Managed Services 

Therefore, teams need to understand **what resources they are using and why**.

10. Cost Allocation

Another important idea is knowing:

Which team, application, or project is responsible for the cost?

For example:

Total Cloud Cost = \$1,000

Team A → \$400

Team B → \$350

Team C → \$250

This makes it easier to identify where money is being spent.

In Kubernetes, labels and namespaces can help organize workloads for monitoring and cost attribution, although the exact cost-allocation approach depends on the tooling being used.

11. Cost Optimization ★

Cost optimization means:

Getting the required performance and reliability while avoiding unnecessary spending.

Common approaches include:

- Right-size CPU and memory.
- Remove unused resources.
- Use autoscaling appropriately.
- Monitor resource utilization.
- Avoid unnecessary Nodes.
- Optimize storage.
- Monitor network usage.
- Choose appropriate cloud resources.
- Set budgets/alerts where supported.

🧠 KCNA Revision Notes

Cost Management

- 💰 Monitor and control cloud/Kubernetes spending.
- 📊 Monitor resource usage.
- 🖥️ Avoid unnecessary Nodes.
- 📁 Avoid unnecessary Pods/resources.
- ⚙️ Use appropriate CPU and memory requests.
- 🚦 Use limits appropriately.
- ✏️ **Right-sizing** = give workloads the resources they actually need.
- 📈 Use autoscaling when appropriate.

-  Track/allocate costs by application, team, or project.
-  Use budgets and cost alerts where available.
-  Remove unused resources.

★ Easy Memory Trick

COST MANAGEMENT

|

├—  Monitor

├—  Right-size

├—  Autoscale

├—  Remove unused resources

└—  Optimize spending

One-line exam definition:

Cost management is the process of monitoring, controlling, and optimizing the cost of cloud and Kubernetes resources while maintaining required performance and reliability.